PRACTICAL 1

AIM:

Project Definition and objective of the specified module and Perform Requirement Engineering Process.

Introduction:

An Online Trading System is a software application that allows users to buy and sell financial instruments through the internet. It replaces traditional manual trading methods with automated and digital processes. The system provides real-time market data, secure transactions, and quick order execution. Users can manage their portfolios and view transaction history easily. This system improves efficiency, accuracy, and transparency in modern trading activities.

Objectives:

- **Primary Objective:**
  - To develop a secure and efficient Online Trading System that enables users to buy and sell financial instruments online.

- **Secondary Objectives:**
  - To provide real-time market data for informed trading decisions.
  - To maintain accurate records of user transactions and portfolio details.

Project Definition:

The Online Trading System is a software-based application designed to facilitate the buying and selling of financial instruments through an internet-based platform. The project focuses on providing a secure, reliable, and user-friendly environment where users can register, access real-time market information, place trading orders, and manage their investment portfolios. The system aims to automate trading operations, reduce manual intervention, and ensure accurate record-keeping while maintaining data security and transparency in all transactions.

Scope:

- Allows users to register and log in securely.
- Provides real-time market price information.
- Enables users to place buy and sell orders online.
- Maintains user portfolio details and transaction history.
- Provides admin control for monitoring users and trading activities.

Modules:

1. **User Management Module:**
   - Handles user registration, login, and authentication.
2. **Trading Module:**
   - Manages buy and sell operations of financial instruments.
3. **Market Data Module:**
   - Provides real-time market prices and trading information.
4. **Portfolio Management Module:**
   - Maintains user investment details and transaction history.

5. **Admin Module:**
   - Controls user activities, system security, and transaction monitor.

Requirement Engineering Process:

- **Stakeholder Identification:** Stakeholders are individuals or groups who have an interest in the system.
  - Primary Stakeholders: Traders, Investors, System Administrators.
  - Secondary Stakeholders: Brokerage Firms, Financial Institutions, Regulatory Authorities.
- **Requirement Elicitation:** This step involves gathering requirements from stakeholders using different techniques.
  - Interviews: Conducting discussions with traders and administrators to understand their needs.
  - Surveys: Collecting feedback from users regarding system features and usability.
  - Observation: Studying existing trading systems to identify limitations and improvements.
- **Requirement Analysis:** In this phase, the collected requirements are analyzed and categorized.
  - **Functional Requirements:**
    1. The system should allow users to register and log in securely.
    2. The system should enable buying and selling of financial instruments.
    3. The system should maintain transaction and portfolio records.

  - **Non-Functional Requirements:**
    1. The system should be secure and reliable.
    2. The system should provide fast response time.
    3. The system should be scalable and user-friendly.

- **Requirement Specification:** All analyzed requirements are documented clearly.
  - Preparation of a Software Requirement Specification (SRS) document.
  - Use case diagrams to represent interactions between users and the system.
  - User stories to describe system behavior from a users perspective.

- **Requirement Validation:** This step ensures that requirements are correct and complete.
  - Reviews: Conducting requirement reviews with stakeholders.
  - Prototyping: Developing basic prototypes to validate functionality.
  - Testing: Verifying that requirements satisfy user expectations.

Lab Activities:

Activity 1: Project Definition

- **Task:** Define the scope and objectives of the Online Trading System.
- **Deliverable:** A detailed project definition document.

Activity 2: Stakeholder Identification

- **Task:** Identify all stakeholders involved in the Online Trading System.
- **Deliverable:** A stakeholder analysis report.

Activity 3: Requirement Elicitation

- **Task:** Gather system requirements using interviews, surveys, and observation.
- **Deliverable:** A requirement elicitation report.

Activity 4: Requirement Analysis

- **Task:** Analyze and classify functional and non-functional requirements.
- **Deliverable:** A requirement analysis document.

Activity 5: Requirement Specification

- **Task:** Prepare the Software Requirement Specification (SRS) document and use case diagrams.
- **Deliverable:** SRS document and use case diagrams.

Activity 6: Requirement Validation

- **Task:** Validate system requirements through reviews and testing.
- **Deliverable:** A requirement validation report.

Conclusion:

This practical demonstrates the application of the requirement engineering process for an Online Trading System. By identifying stakeholders and systematically gathering, analyzing, and validating requirements, the system can be developed efficiently. Proper requirement engineering ensures the development of a secure, reliable, and user-friendly trading platform that meets user and business needs.